

Chapter 28

■ Risk Analysis

Slide Set to accompany

Software Engineering: A Practitioner's Approach, 7/e

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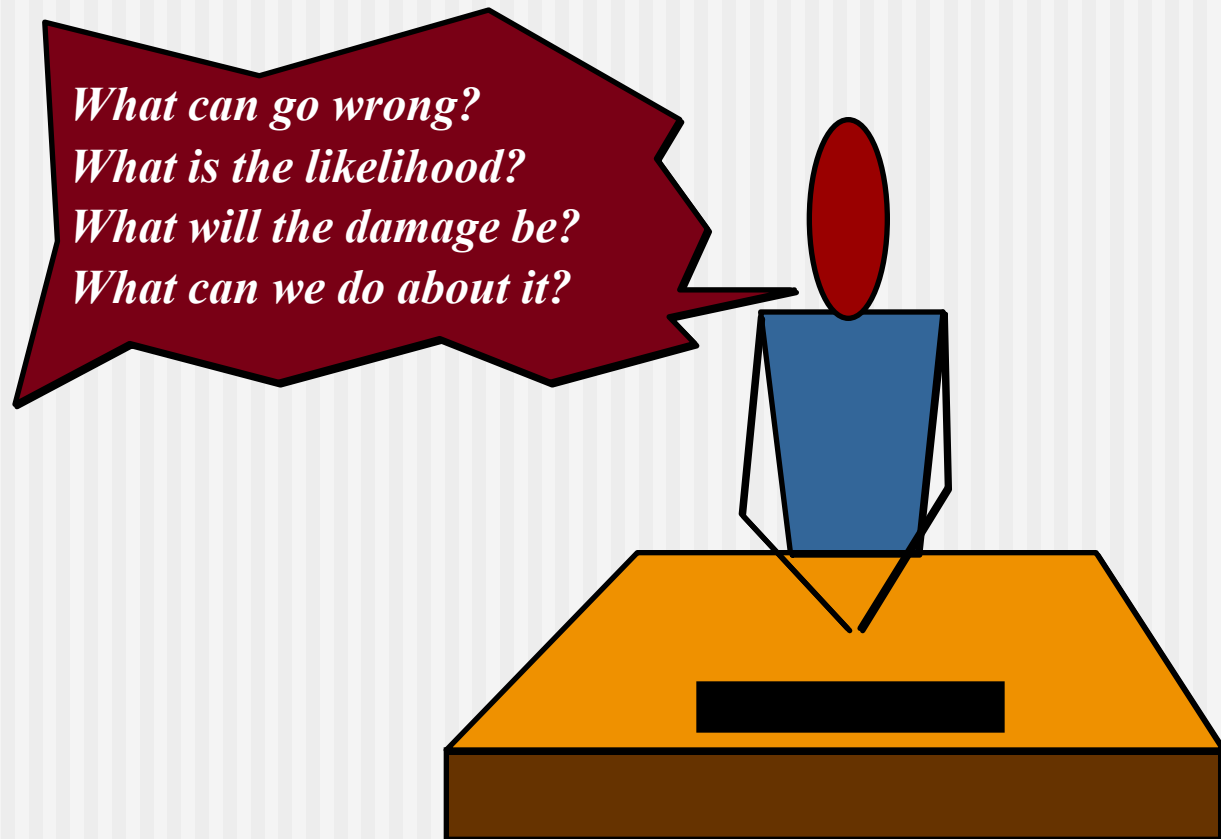
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Project Risks



Definition of Risk

- A risk is a potential problem – it might happen and it might not
- Conceptual definition of risk
 - Risk concerns future happenings
 - Risk involves change in mind, opinion, actions, places, etc.
 - Risk involves choice and the uncertainty that choice entails
- Two characteristics of risk
 - Uncertainty – the risk may or may not happen, that is, there are no 100% risks (those, instead, are called constraints)
 - Loss – the risk becomes a reality and unwanted consequences or losses occur

Software Risk

Risk Categorization – Approach #1

- Project risks
 - They threaten the project plan
 - If they become real, it is likely that the project schedule will slip and that costs will increase
- Technical risks
 - They threaten the quality and timeliness of the software to be produced
 - If they become real, implementation may become difficult or impossible
- Business risks
 - They threaten the viability of the software to be built
 - If they become real, they jeopardize the project or the product

Software Risk

- Sub-categories of Business risks
 - **Market risk** – building an excellent product or system that no one really wants
 - **Strategic risk** – building a product that no longer fits into the overall business strategy for the company
 - **Sales risk** – building a product that the sales force doesn't understand how to sell
 - **Management risk** – losing the support of senior management due to a change in focus or a change in people
 - **Budget risk** – losing budgetary or personnel commitment
- Known risks
 - Those risks that can be uncovered after careful evaluation of the project plan, the business and technical environment in which the project is being developed, and other reliable information sources (e.g., unrealistic delivery date)
- Predictable risks
 - Those risks that are extrapolated from past project experience (e.g., past turnover)
- Unpredictable risks
 - Those risks that can and do occur, but are extremely difficult to identify in advance

Reactive Risk Management

- project team reacts to risks when they occur
- mitigation—plan for additional resources in anticipation of fire fighting
- fix on failure—resources are found and applied when the risk strikes
- crisis management—failure does not respond to applied resources and project is in jeopardy

Proactive Risk Management

- formal risk analysis is performed
- organization corrects the root causes of risk
 - TQM (Total Quality Management) concepts and statistical SQA (Software Quality Assurance)
 - examining risk sources that lie beyond the bounds of the software
 - developing the skill to manage change

Steps for Risk Management

- 1) Identify possible risks; recognize what can go wrong
- 2) Analyze each risk to estimate the probability that it will occur and the impact (i.e., damage) that it will do if it does occur
- 3) Rank the risks by probability and impact
 - Impact may be negligible, marginal, critical, and catastrophic
- 4) Develop a contingency plan to manage those risks having high probability and high impact

Seven Principles of Risk Management

- **Maintain a global perspective**—view software risks within the context of system and the business problem
- **Take a forward-looking view**—think about the risks that may arise in the future; establish contingency plans
- **Encourage open communication**—if someone states a potential risk, don't discount it.
- **Integrate**—a consideration of risk must be integrated into the software process
- **Emphasize a continuous process**—the team must be vigilant throughout the software process, modifying identified risks as more information is known and adding new ones as better insight is achieved.
- **Develop a shared product vision**—if all stakeholders share the same vision of the software, it likely that better risk identification and assessment will occur.
- **Encourage teamwork**—the talents, skills and knowledge of all stakeholder should be pooled

Risk Management Paradigm



Risk Identification

- *Product size*—risks associated with the overall size of the software to be built or modified.
- *Business impact*—risks associated with constraints imposed by management or the marketplace.
- *Customer characteristics*—risks associated with the sophistication of the customer and the developer's ability to communicate with the customer in a timely manner.
- *Process definition*—risks associated with the degree to which the software process has been defined and is followed by the development organization.
- *Development environment*—risks associated with the availability and quality of the tools to be used to build the product.
- *Technology to be built*—risks associated with the complexity of the system to be built and the "newness" of the technology that is packaged by the system.
- *Staff size and experience*—risks associated with the overall technical and project experience of the software engineers who will do the work.

Assessing Project Risk-I

- Have top software and customer managers formally committed to support the project?
- Are end-users enthusiastically committed to the project and the system/product to be built?
- Are requirements fully understood by the software engineering team and their customers?
- Have customers been involved fully in the definition of requirements?
- Do end-users have realistic expectations?

Assessing Project Risk-II

- Is project scope stable?
- Does the software engineering team have the right mix of skills?
- Are project requirements stable?
- Does the project team have experience with the technology to be implemented?
- Is the number of people on the project team adequate to do the job?
- Do all customer/user constituencies agree on the importance of the project and on the requirements for the system/product to be built?

Risk Components

- *performance risk*—the degree of uncertainty that the product will meet its requirements and be fit for its intended use.
- *cost risk*—the degree of uncertainty that the project budget will be maintained.
- *support risk*—the degree of uncertainty that the resultant software will be easy to correct, adapt, and enhance.
- *schedule risk*—the degree of uncertainty that the project schedule will be maintained and that the product will be delivered on time.

Risk Projection

- *Risk projection*, also called *risk estimation*, attempts to rate each risk in two ways
 - the likelihood or probability that the risk is real
 - the consequences of the problems associated with the risk, should it occur.
- There are four risk projection steps:
 - establish a scale that reflects the perceived likelihood of a risk
 - delineate the consequences of the risk
 - estimate the impact of the risk on the project and the product,
 - note the overall accuracy of the risk projection so that there will be no misunderstandings.

Building a Risk Table

Risk	Probability	Impact	RMM M
			Risk Mitigation Monitoring & Management

Building the Risk Table

- Estimate the **probability** of occurrence
- Estimate the **impact** on the project on a scale of 1 to 5, where
 - 1 = low impact on project success
 - 5 = catastrophic impact on project success
- sort the table by probability and impact

Risk Exposure (Impact)

The overall *risk exposure*, **RE**, is determined using the following relationship [Hal98]:

$$\mathbf{RE} = \mathbf{P} \times \mathbf{C}$$

where

P is the probability of occurrence for a risk, and
C is the cost to the project should the risk occur.

Risk Exposure Example

- **Risk identification.** Only 70 percent of the software components scheduled for reuse will, in fact, be integrated into the application. The remaining functionality will have to be custom developed.
- **Risk probability.** 80% (likely).
- **Risk impact.** 60 reusable software components were planned. If only 70 percent can be used, 18 components would have to be developed from scratch (in addition to other custom software that has been scheduled for development). Since the average component is 100 LOC and local data indicate that the software engineering cost for each LOC is \$14.00, the overall cost (impact) to develop the components would be $18 \times 100 \times 14 = \$25,200$.
- **Risk exposure.** $RE = 0.80 \times 25,200 \sim \$20,200$.

Risk Mitigation, Monitoring, and Management

- **mitigation**—how can we avoid the risk?
- **monitoring**—what factors can we track that will enable us to determine if the risk is becoming more or less likely?
- **management**—what contingency plans do we have if the risk becomes a reality?

Risk information sheet			
Risk ID: P02-4-32	Date: 5/9/09	Prob: 80%	Impact: high
Description: Only 70 percent of the software components scheduled for reuse will, in fact, be integrated into the application. The remaining functionality will have to be custom developed.			
Refinement/context: Subcondition 1: Certain reusable components were developed by a third party with no knowledge of internal design standards. Subcondition 2: The design standard for component interfaces has not been solidified and may not conform to certain existing reusable components. Subcondition 3: Certain reusable components have been implemented in a language that is not supported on the target environment.			
Mitigation/monitoring: 1. Contact third party to determine conformance with design standards. 2. Press for interface standards completion; consider component structure when deciding on interface protocol. 3. Check to determine number of components in subcondition 3 category; check to determine if language support can be acquired.			
Management/contingency plan/trigger: RE computed to be \$20,200. Allocate this amount within project contingency cost. Develop revised schedule assuming that 18 additional components will have to be custom built; allocate staff accordingly. Trigger: Mitigation steps unproductive as of 7/1/09.			
Current status: 5/12/09: Mitigation steps initiated.			
Originator: D. Gagne		Assigned: B. Laster	

Risk Due to Product Size

Attributes that affect risk:

- **estimated size of the product in LOC or FP?**
- **estimated size of product in number of programs, files, transactions?**
- **percentage deviation in size of product from average for previous products?**
- **size of database created or used by the product?**
- **number of users of the product?**
- **number of projected changes to the requirements for the product? before delivery? after delivery?**
- **amount of reused software?**

Risk Due to Business Impact

Attributes that affect risk:

- **affect of this product on company revenue?**
- **visibility of this product by senior management?**
- **reasonableness of delivery deadline?**
- **number of customers who will use this product**
- **interoperability constraints**
- **sophistication of end users?**
- **amount and quality of product documentation that must be produced and delivered to the customer?**
- **governmental constraints**
- **costs associated with late delivery?**
- **costs associated with a defective product?**

Risks Due to the Customer

Questions that must be answered:

- **Have you worked with the customer in the past?**
- **Does the customer have a solid idea of requirements?**
- **Has the customer agreed to spend time with you?**
- **Is the customer willing to participate in reviews?**
- **Is the customer technically sophisticated?**
- **Is the customer willing to let your people do their job—that is, will the customer resist looking over your shoulder during technically detailed work?**
- **Does the customer understand the software engineering process?**

Risks Due to Process Maturity

Questions that must be answered:

- **Have you established a common process framework?**
- **Is it followed by project teams?**
- **Do you have management support for software engineering**
- **Do you have a proactive approach to SQA?**
- **Do you conduct formal technical reviews?**
- **Are CASE tools used for analysis, design and testing?**
- **Are the tools integrated with one another?**
- **Have document formats been established?**

Technology Risks

Questions that must be answered:

- **Is the technology new to your organization?**
- **Are new algorithms, I/O technology required?**
- **Is new or unproven hardware involved?**
- **Does the application interface with new software?**
- **Is a specialized user interface required?**
- **Is the application radically different?**
- **Are you using new software engineering methods?**
- **Are you using unconventional software development methods, such as formal methods, AI-based approaches, artificial neural networks?**
- **Are there significant performance constraints?**
- **Is there doubt the functionality requested is "do-able?"**

Staff/People Risks

Questions that must be answered:

- **Are the best people available?**
- **Does staff have the right skills?**
- **Are enough people available?**
- **Are staff committed for entire duration?**
- **Will some people work part time?**
- **Do staff have the right expectations?**
- **Have staff received necessary training?**
- **Will turnover among staff be low?**

Recording Risk Information

Project: Embedded software for XYZ system

Risk type: schedule risk

Priority (1 low ... 5 critical): 4

Risk factor: Project completion will depend on tests which require hardware component under development. Hardware component delivery may be delayed

Probability: 60 %

Impact: Project completion will be delayed for each day that hardware is unavailable for use in software testing

Monitoring approach:

Scheduled milestone reviews with hardware group

Contingency plan:

Modification of testing strategy to accommodate delay using software simulation

Estimated resources: 6 additional person months beginning in July